Troy Mayer

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EDUCATION

Purdue University, School of Mechanical Engineering and School of Science — West Lafayette, IN
B.S. Mechanical Engineering; B.S. Applied Statistics; Minor in Economics — May 2028
GPA 3.32/4 | Clubs: Club Swimming, House Band

PROFESSIONAL EXPERIENCE

Boston Dynamics — Waltham, MA
Sensors and Perception Co-Op, Atlas — Jul 2026 – Present

- Computing limb position and velocity at impact with floating-base forward kinematics (quaternions, Rodrigues' rotations, rigid-body velocity composition) from joint encoders, and mapping impacts onto the robot's mesh.
- Estimating impact energy from Khatib's operational-space effective mass and an OLS velocity fit, and building a physics-based MuJoCo impact simulation.
- Built an impact forensics pipeline and Flask/Plotly dashboard on a Linux VM covering ~28k impact events from ~900 robot test logs, with ETL from a time-series telemetry store into SQLite, a 3D impact-location heatmap, and per-event waveform inspection used by engineers during testing.
- Modeling hardware thermal behavior in Python (Pandas, SciPy), fitting heating curves by multi-start least squares to predict steady-state temperatures across hundreds of units without full-length tests.

Caterpillar (via Purdue Data Mine) — West Lafayette, IN
Machine Learning Researcher — Jan 2026 – May 2026

- Built LSTM and XGBoost models forecasting next-day primer, topcoat, and solvent consumption from five years of paint-line data, with Monte Carlo confidence intervals on each forecast, achieving a 4.7% reduction in material usage.
- Built Python pipelines consolidating five years of noisy legacy data into one daily series, contributing to a 3.3% reduction in cycle time.

Chortos Lab, Soft Adaptive Systems and Structures — West Lafayette, IN
Undergraduate Researcher — Aug 2025 – May 2026

- Wrote a Python computer-vision tracker resolving cantilever displacement to 0.01 mm and built LabVIEW/MATLAB data acquisition for 50,000+ point datasets, guiding gel formulations that increased force output by over 50%.

PROJECTS

Legged Robot State Estimation and Control | *Personal Project* — Aug 2026 – Oct 2026

- Built a contact-aided error-state EKF in C++/Eigen fusing a 1 kHz IMU with leg kinematics on a simulated Unitree Go1, reaching 0.013 m/s velocity RMSE over a 60 s walk vs. 1.38 m/s IMU-only, at ~20 µs per update.
- Verified Jacobians against finite differences in GoogleTest, matched a Python reference to 1e-13, and showed analytically and numerically that roll, pitch, and velocity stay bounded while position and yaw drift.
- Ran a trot controller on the estimate for 5 × 60 s walks with 0 falls, tracking within 0.004 m/s of the ground-truth-fed run; added a 500 Hz QP force-balance controller that cut ground-truth velocity-tracking error from 0.169 to 0.091 m/s.

Active Orthosis, Purdue ASME Biomechanics | *Electronics Team Lead* — Apr 2024 – May 2026

- Designed and implemented a 1 kHz PI velocity controller in C++ on an ESP32-S3, tuned from a first-order motor model with feed-forward and anti-windup (0.8% speed error lifting the forearm in simulation), protected by 10 fault monitors, a hardware watchdog, and a kill switch.
- Cut EMG-to-motion latency from 103 ms to 43 ms with a Bayesian EMG estimator, validated in closed-loop simulation on real recorded EMG with 26 GoogleTest tests in CI.
- Led a 5-engineer team and redesigned the 4-channel EMG board in KiCad with an INA333 front end, right-leg drive, and ISO7763 motor isolation, raising cross-day gesture decoding from 31% to 87% in emulation.

Asteroid Hopping Robot | *Independent Research, Purdue University & UC3M (Madrid)* — Oct 2026 – Present

- Designing push-off control optimized to hit a target launch speed, knee impedance control for landing, and reaction-wheel control to reorient mid-air for a three-legged robot that changes leg shape between hops.
- Simulating hops in MuJoCo, comparing risk-limited rules that trade distance against the chance of escaping the asteroid or tipping over, and building a pendulum rig to test landings.

SKILLS

- **Programming:** C++17, Python, MATLAB, SQL | Languages: English, Turkish (fluent), Spanish (fluent)
- **Robotics and controls:** Kalman filtering (error-state EKF), sensor fusion, rigid-body kinematics, PI and feed-forward control, QP force control, MuJoCo, Eigen, FreeRTOS, ESP32
- **Software and data:** Linux, Git, GitHub Actions CI, CMake, GoogleTest, Flask, Plotly, SQLite, Pandas, SciPy, PyTorch, XGBoost, Snowflake
- **Hardware and CAD:** KiCad, LabVIEW, DAQ, CATIA, NX, Fusion 360, Creo, FEA